L-P-397C
December 16, 1977
SUPERSEDING
Fed. Spec. L-P-397B
November 30, 1971

FEDERAL SPECIFICATION

PLASTIC MOLDING MATERIAL, CELLULOSE ACETATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers certain general purpose plasticized cellulose acetate compounds suitable for injection molding (see 6.1).

1.2 Classification. Cellulose acetate compounds covered by this specification shall be of six grades designated H4, H3, H2, H, MH, and M in order of decreasing flow temperature, tensile strength, but increasing impact strength as specified in table I.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-D-723 - Drums, Fiber.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55 Gallon

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 9330

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Philadelphia, Atlanta, Chicago, Kansas City, MO., Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 256 - Impact Resistance of Plastics and Electrical Insulating Materials.
D 569 - Measuring the Flow Properties of Thermoplastic Molding Materials.
D 570 - Water Absorption of Plastics.
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing.
D 635 - Flammability of Self-Supporting Plastics.
D 638 --Tensile Properties of Plastics.
D 706 - Cellulose Acetate Molding and Extrusion Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations Inc., Traffic Department, 1616 P Street, N. W., Washington DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The molding compounds shall consist of cellulose acetate polymer with suitable added plasticizers and with or without added dyes and pigments. Unless otherwise specified, the compounds shall be suitable for injection molding (see 6.1 and 6.2).

3.2 Property values. The material shall conform to the property values specified in table I, when tested in accordance with the applicable procedure of 4.3.

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TABLE I. Property values[1/]

Property[1/]	Values					
	Grade H4	Grade H3	Grade H2	Grade H	Grade MH	Grade M
Flow temperature, °C ± 5°C (°F ± 9°F)	170 (338)	165 (329)	160 (320)	155 (311)	150 (302)	145 (293)
Tensile strength, MPa (psi), min.	41.4 (6000)	40.0 (5800)	37.9 (5500)	33.1 (4800)	29.0 (4200)	24.1 (3500)
Impact strength, Izod, J/m of notch (ft. lbs. per inch of notch), min.	53.4 (1.0)	85.4 (1.6)	123 (2.3)	160 (3.0)	198 (3.7)	235 (4.4)
Weight loss on heating, percent, max.	1.0	2.0	2.5	3.0	4.0	6.5
Water absorption						
Weight gain plus soluble matter loss, percent, max.	2.5	2.5	2.8	3.0	3.5	4.0
Soluble matter loss, percent, max.	0.3	0.3	0.4	0.5	0.6	0.8
Flammability						
Average extent of burning, mm (inch) per minute, max.	38.1 (1.5)	38.1 (1.5)	38.1 (1.5)	38.1 (1.5)	38.1 (1.5)	38.1 (1.5)

[1/] Metric unit abbreviations: MPa = megapascal, J/m = joules per meter, mm = millimetre.

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3.3 Color. The material shall be of the color and transparency or opacity specified by the procuring activity, as determined on plaques molded and tested as specified.

3.4 Form. Unless otherwise specified by the procuring activity, the material shall be supplied in the form of pellets of the size and shape normally furnished by the supplier.

3.5 Uniformity of color and form. The pellets, or other form as specified by the procuring activity, shall be uniform in color and form within each container and from container to container.

3.6 Suitability (compatibility) for use with explosives. (Applicable only when specified by the procuring activity.) When so specified, application shall be made to a Government laboratory or other Government approved laboratory designated by the procuring activity for determination of suitability of the material for use with a particular explosive or explosives (see 6.2).

3.7 Workmanship. The material shall be clean, uniform in appearance, and free from contamination as determined by visual examination.

3.8 Recovered materials. The material offered shall contain not less than 0.1 percent of recovered materials. Recovered materials are defined as material which has been collected or recovered from solid waste. Solid waste is defined as any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility and other discarded material including solid, liquid, semisolid, or contained gaseous material resulting from community activities. Excluded are solid or dissolved material in domestic sewage, or solid or dissolved materials in irrigation return flows or specific industrial discharges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, form and color submitted for delivery at one time.

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4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, inspected, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced specifications and standards. In the event of conflict this specification shall govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of material. Examination of material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 45.4 kilograms (100 pounds) for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of material for defects in color, form, appearance, and workmanship. The sample unit for this examination, specified in table II shall be approximately 0.454 kilograms (1 pound).

TABLE II. Examination for defects in color, form, appearance and workmanship

Examine	Defect
Color	Color not as specified. Color not uniform.
Form	Form not as specified. Form not uniform.
Appearance and workmanship	Not clean, not free of contamination, not uniform in appearance.

4.2.2.1.2 Examination of preparation for delivery. An examination shall be made in accordance with table III to determine that packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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TABLE III. Examination of preparation for delivery

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements.
	Nonconforming component, component missing, damaged or otherwise defective affecting serviceability.
	Inadequate application of components such as incomplete closure of polyethylene liners.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

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4.2.3 Classification of tests. All tests shall be classified as follows:

a. Lot acceptance tests (see 4.2.3.1).
b. Periodic lot check tests (see 4.2.3.2).

4.2.3.1 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot except when periodic lot check tests are required. The lot acceptance tests shall be as specified in table IV.

TABLE IV. Lot acceptance tests

Test
Flow temperature
Tensile strength
Impact strength

4.2.3.2 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring activity (see 6.2). Periodic lot check tests shall consist of all tests specified in table I plus the test for color. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot. Unless otherwise specified by the procuring activity, periodic lot check test results shall be included with each lot requiring periodic lot check tests.

4.2.4 Testing. The material shall be tested for the applicable characteristics specified in section 3 in accordance with the test methods specified for each lot submitted for inspection. Sampling shall be in accordance with MIL-STD-105, and the lot size for the purpose of determining sample size for testing shall be expressed in units of 90.7 kilograms (200 pounds) of material. The inspection level shall be S-1. The sample size (number of units of product drawn from a lot) shall be in accordance with the single sampling plan for normal inspection, and each unit shall consist of sufficient material to prepare all specimens required for testing. Each sample unit shall be tested and must meet all applicable requirements of section 3. Test results for each characteristic in each sample unit shall be the averaged results from the specimens tested for that characteristic.

4.3 Test methods.

4.3.1 Specimen preparation. Test specimens shall be prepared by injection molding in accordance with specific conditions recommended by the manufacturer. If recommendations by the manufacturer are not available, molding shall be in accordance with ASTM D 706.

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4.3.2 Specimen conditioning. Specimens shall be conditioned in accordance with procedure A of ASTM D 618 except that flow temperature, weight loss on heating, and water absorption specimens shall be conditioned as specified in the applicable test method.

4.3.3 Specimen testing. Specimens shall be tested at 23° ± 2°C (73.4 ± 3.6°F) and 50 ± 5 percent relative humidity, except where otherwise specified in the applicable test method.

4.3.4 Flow temperature. Three specimens shall be tested in accordance with procedure A of ASTM D 569. Conditioning immediately prior to testing and dimensions of test specimens shall be in accordance with that specified in ASTM D 569.

4.3.5 Tensile strength. Three specimens shall be tested in accordance with ASTM D 638 at speed B and using the dimensions of type I specimens.

4.3.6 Impact strength. Three specimens shall be tested in accordance with method A of ASTM D 256. Specimen dimensions shall be 63.5mm (2.5 inches) long by 12.7mm (0.5 inch) wide by 3.2mm (0.125 inch) thick.

4.3.7 Weight loss on heating. Three test specimens each 50.8mm (2 inches) in diameter by 3.2mm (0.125 inch) thick or 50.8mm (2 inches) wide by 63.5mm (2.5 inches) long by 3.2mm (0.125 inch) thick shall be conditioned for a minimum of 48 hours over anhydrous calcium chloride at 23° ± 2°C (73.4° ± 3.6°F). The three test specimens shall be weighed and then placed flat on a screen in a circulating air oven for 72 ± 1 hours at 82° ± 2°C (180° ± 3.6°F). Upon removal from the oven, test specimens shall be cooled in a desiccator over anhydrous calcium chloride to 23° ± 2°C (73.4° ± 3.6°F). The specimens shall be removed from the desiccator and weighed. The percentage weight loss shall be calculated on the basis of the original conditioned weight.

4.3.8 Water absorption. Three specimens shall be tested in accordance with ASTM D 570. Specimen conditioning shall be as specified in ASTM D 570.

4.3.9 Flammability. Ten specimens shall be tested in accordance with ASTM D 635. Specimen dimensions shall be as specified in ASTM D 635, and the average extent of burning shall be determined.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.1.1 Level A. Unless otherwise specified, the material shall be packed in metal drums conforming to PPP-D-729, type III or type IV, in quantities of 181 kilograms (400 pounds) maximum.

Insofar as practical, containers shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Containers shall be filled to identical quantities and shall be closed in accordance with the applicable container specification.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers.

a. Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 90.7 kilograms (200 pounds) maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 181 kilograms (400 pounds) maximum.

Insofar as practical, containers shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Containers shall be filled to identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.102 millimetre (0.004 inch) thick polyethylene liner properly heat sealed.

5.1.3 Commercial. Packing shall be in accordance with the commercial practice adequate to ensure acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Compounds covered by this specification are intended primarily for fabrication into parts by the injection molding process. Although some of the compounds also may be suitable for extrusion, the manufacturer should be consulted before they are ordered for that purpose. Extrusion compounds usually are of somewhat different composition. The chief characteristics of cellulose acetate molded components are their toughness, high impact strength, and ease of fabrication. In designating molded articles of cellulose acetate plastics, their limited resistance to cold flow and the dimensional changes to be expected in the finished item should be considered. Cellulose acetate plastics are suitable for applications in which slight shrinkage on aging can be tolerated and involving relatively light loads. The optical properties of these plastics are good, as are their electrical characteristics, though the latter deteriorate rapidly under humid conditions (and it should be noted that there are some newer plastics available with better electrical properties) (see MIL-HDBK-700(MR)).

6.1.1 Drying. Cellulose acetate absorbs small amounts of water and should be dried before it is processed. The recommended dry time is generally between 1 and 3 hours in a low-humidity dryer at 71°C (160°F) or above.

6.1.2 Cooling. The molded part must be cooled slowly. If it is cooled too rapidly, a hard skin will form on the surface and become highly stressed as the inside of the part cools and shrinks. This results in frozen-in-stresses, which detract from toughness and can also cause voids to form in the interior of thick sections. Optimum cooling conditions are attained when the hot part is passed through three water troughs containing in order hot water, warm water, and cold water.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade of material required (see 1.2 and table I).
(c) Color and transparency required, with tolerance and method of measurement (see 3.3 and 6.3).
(d) Form, if required (see 3.4).
(e) Periodic lot check tests when required (see 4.2.3.2).
(f) Degree of packing, if required (see 5.1).
(g) Suitability for use with explosives and method of determination, if required (see 3.6).

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6.3 Determination of color. When a particular color or transparency is required, a standard sample accompanied by borderline standards or a written statement defining the permissible tolerances and method of determination may be furnished by the procuring agency or furnished by the supplier and approved by the procuring agency. Since the color of a molded item may depend on its thickness and method of molding, and since cellulose acetate items usually are produced by injection molding, certain applications may require color matching on injection molded plaques of specified thickness.

6.3.1 Mixtures. When a mottled effect, to be obtained by mixing two materials of different colors in the injection molding machine, is required, the grade and color of each component and the ratio of mixing the two required to produce a molding of the required color and other properties must be determined by experiment. Then the grade and color of each component must be specified separately, and the molder is responsible for the color and properties of the final item.

6.4 Supersession data. The following table lists the grades of this specification corresponding to the types and classes of superseded specification L-P-397A. Not all values for corresponding designations are the same.

Classification in L-P-397C and L-P-397B	Classification in L-P-397A
Grade H4	Type I, class 1
Grade H3	Type I, class 2
Grade H2	Type I, class 3
Grade H	Type I, class 4
Grade MH	Type I, class 5
Grade M	Type I, class 6

MILITARY CUSTODIANS:

Army - MR
Navy - SH
Air Force - 11
DLA - GS

Review activities:

Army - EL, AR, MI, SM, ME, GL
Navy - SH
Air Force - 11, 17, 99

Preparing activity:

Army - MR

CIVIL AGENCY COORDINATING ACTIVITIES:

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COMMERCE - NBS
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